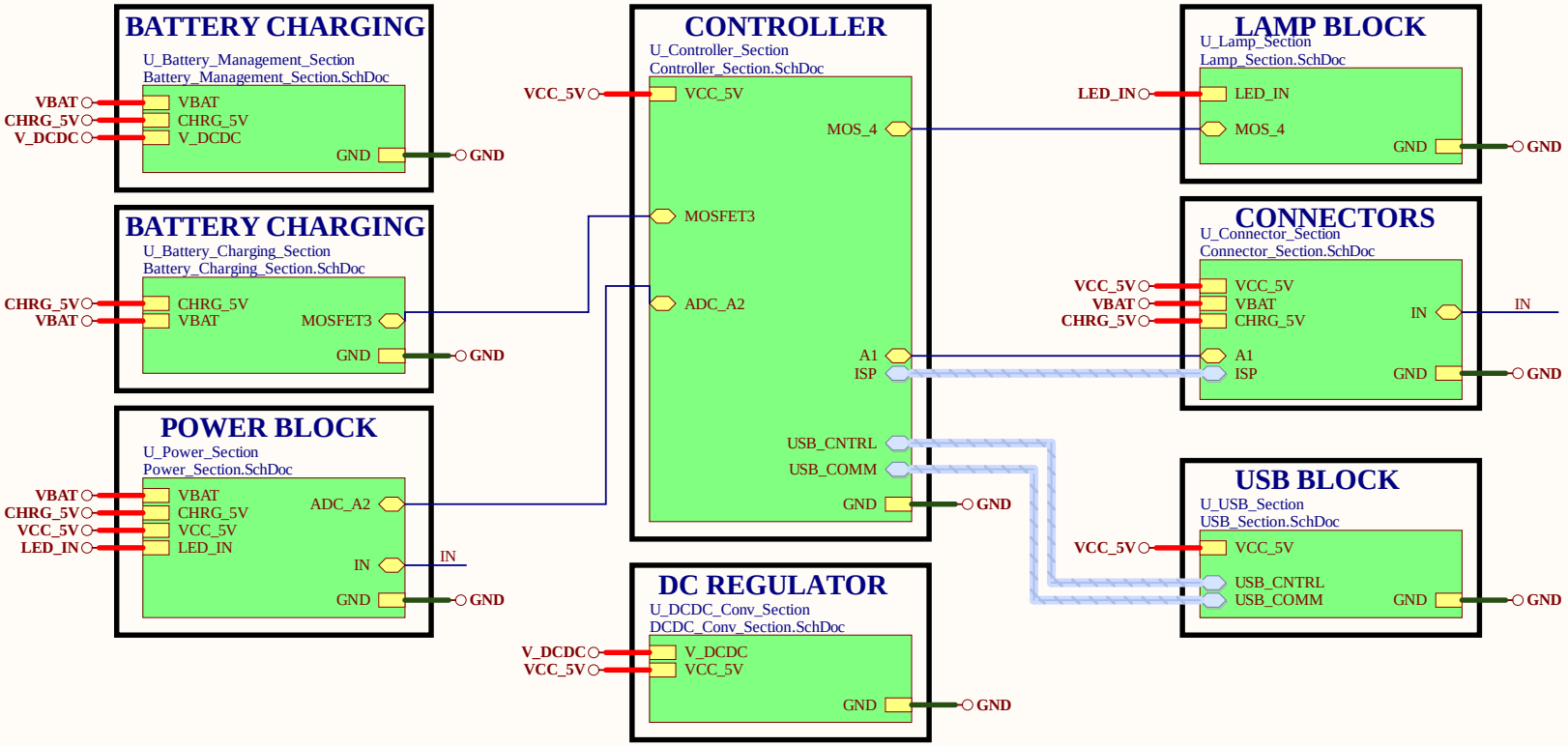
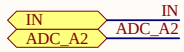
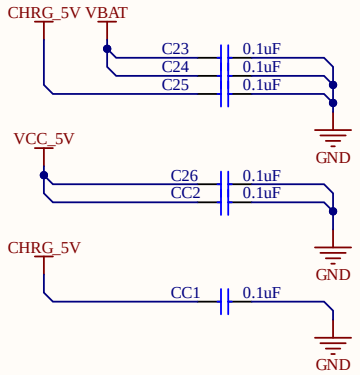
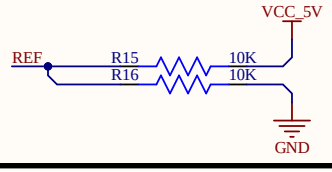

BATTERY CHARGING
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT
CHRG_5V
V_DCDC
GND
BATTERY CHARGING
U_Battery_Charging_Section
Battery_Charging_Section.SchDoc
CHRG_5V
VBAT
MOSFET3
GND
POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VBAT
CHRG_5V
VCC_5V
LED_IN
ADC_A2
IN
GND
CONTROLLER
U_Controller_Section
Controller_Section.SchDoc
VCC_5V
MOS_4
MOSFET3
ADC_A2
A1
ISP
USB_CNTRL
USB_COMM
GND
DC REGULATOR
U_DCDC_Conv_Section
DCDC_Conv_Section.SchDoc
V_DCDC
VCC_5V
GND
LAMP BLOCK
U_Lamp_Section
Lamp_Section.SchDoc
LED_IN
MOS_4
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VCC_5V
VBAT
CHRG_5V
IN
A1
ISP
GND
USB BLOCK
U_USB_Section
USB_Section.SchDoc
VCC_5V
USB_CNTRL
USB_COMM
GND



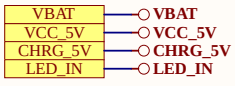
DECOUPLING CAPACITOR



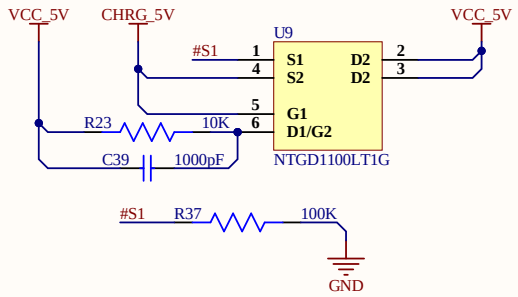
RESISTOR DIVIDER



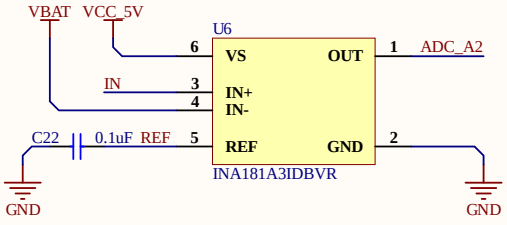
POWER SIGNALS



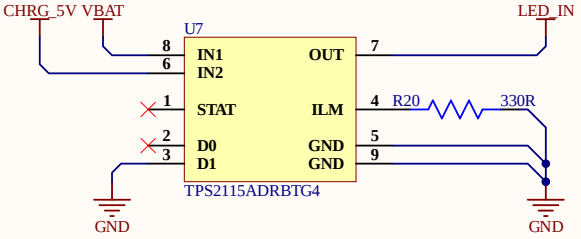
POWER DRIVING IC



CURRENT SENSE AMPLIFIER



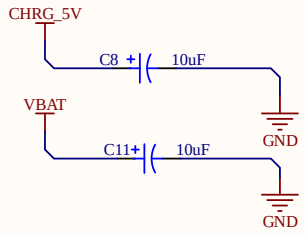
SOURCE SELECTOR



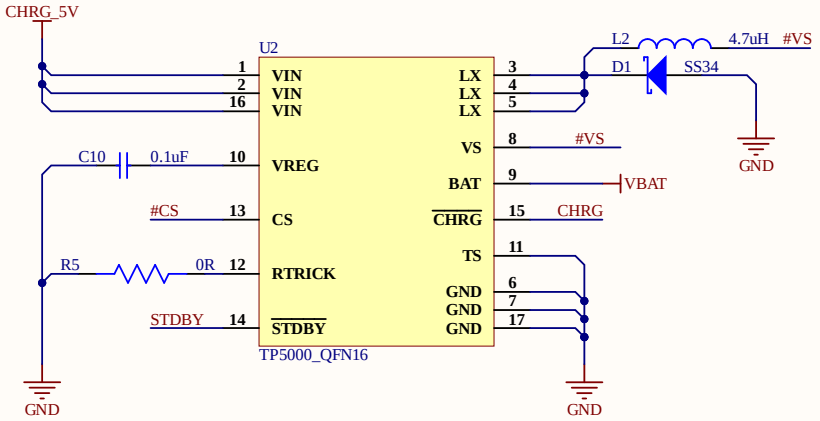
STAT pin can be pulled up to output and connected to MCU for getting IC status

IL=500/RILM

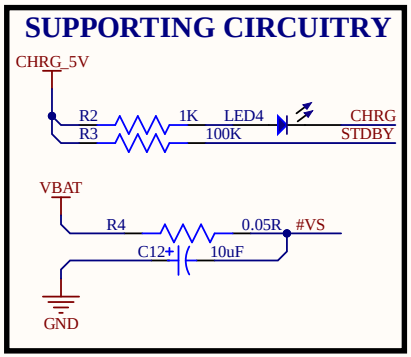
DECOUPLING CAPACITORS



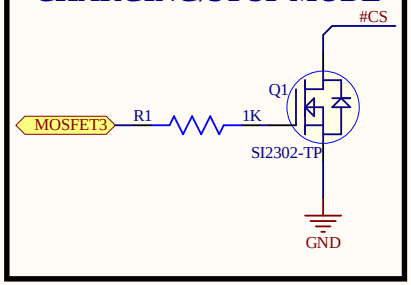
BATTERY CHARGING IC



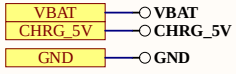
SUPPORTING CIRCUITRY



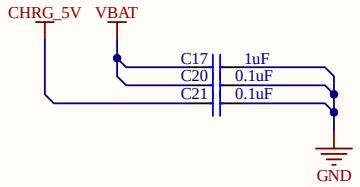
CONTROL CIRCUIT FOR CHARGING/STOP MODE



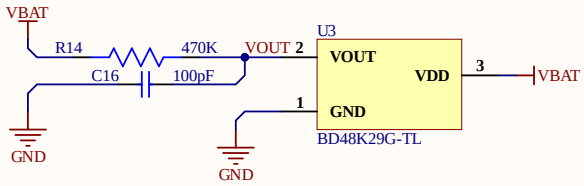
POWER SIGNALS



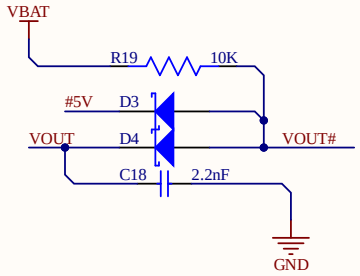
DECOUPLING CAPACITOR



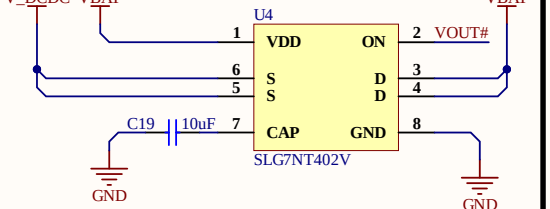
BATTERY SUPERVISORY IC



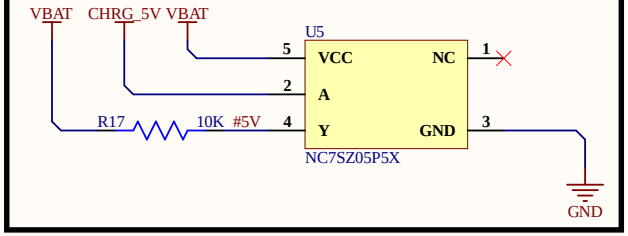
SUPPORTING CIRCUITRY



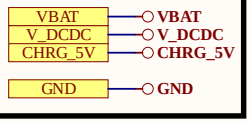
POWER SWITCH



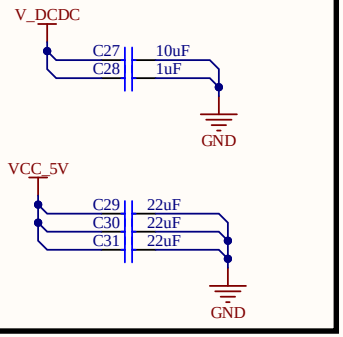
BUFFER INVERTER



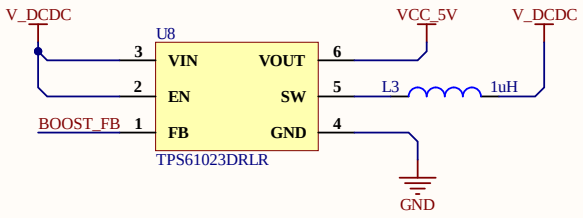
POWER SIGNALS



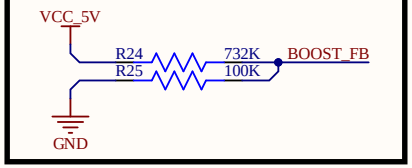
DECOUPLING CAPACITOR



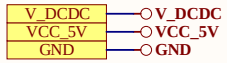
5V BOOST REGULATOR

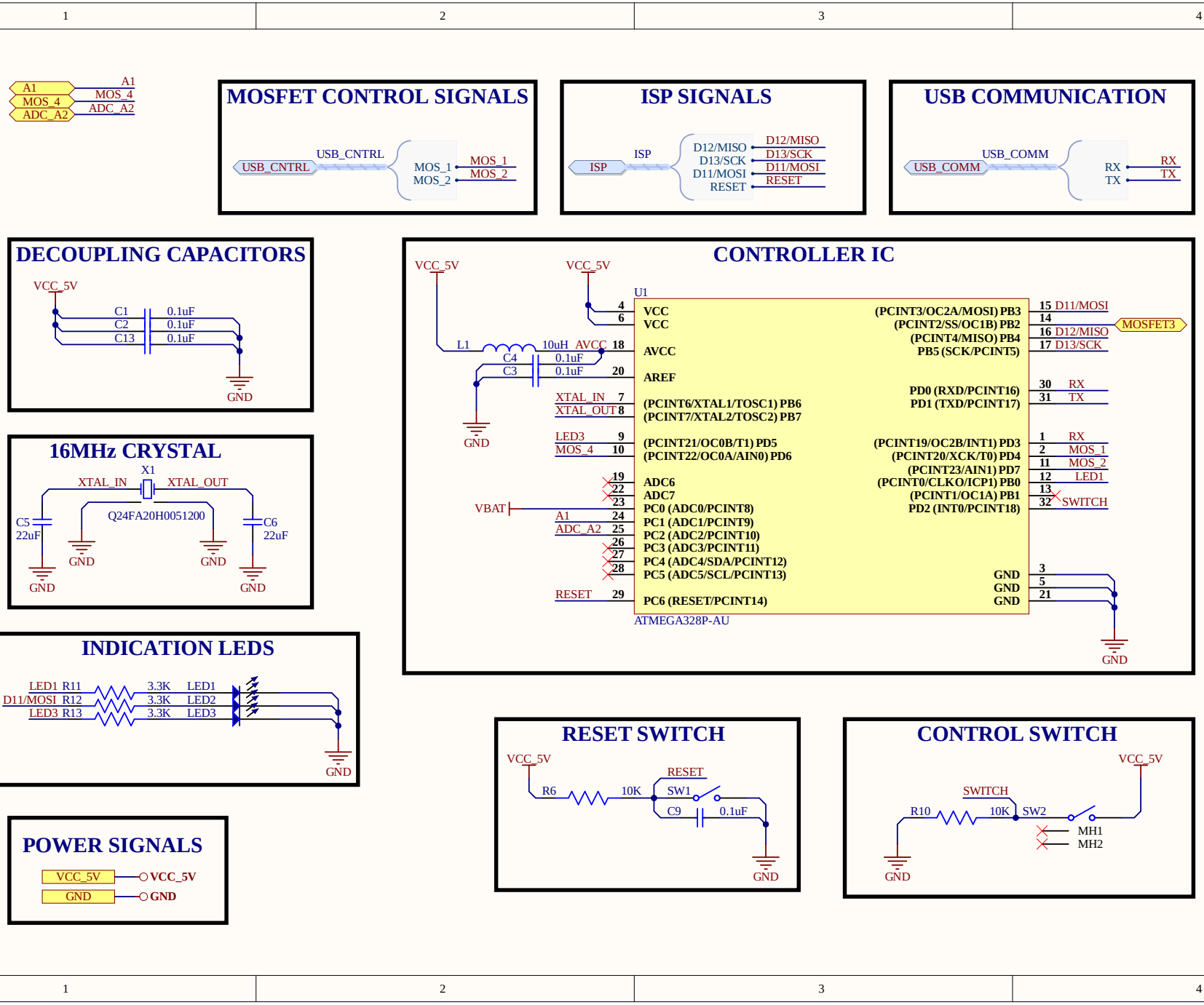


RESISTOR DIVIDER



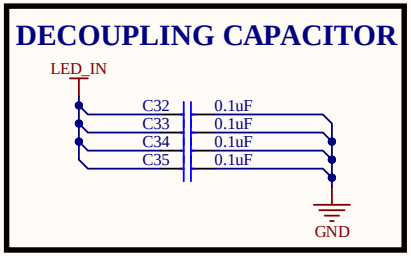
POWER SIGNALS



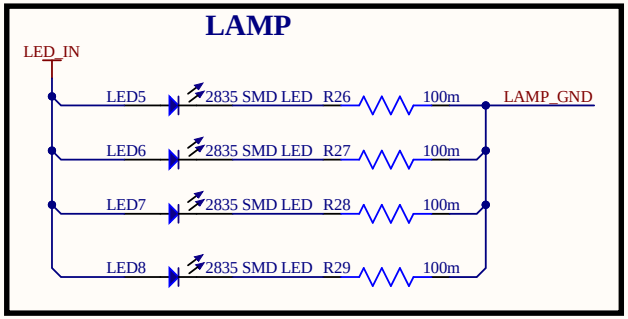

A1
MOS_4
ADC_A2
MOSFET CONTROL SIGNALS
USB_CNTRL
MOS_1
MOS_2
ISP SIGNALS
ISP
D12/MISO
D13/SCK
D11/MOSI
RESET
USB COMMUNICATION
USB_COMM
RX
TX
DECOUPLING CAPACITORS
VCC_5V
C1 0.1uF
C2 0.1uF
C13 0.1uF
GND
16MHz CRYSTAL
X1
XTAL_IN
XTAL_OUT
Q24FA20H0051200
C5 22uF
C6 22uF
CONTROLLER IC
U1
VCC
AVCC
AREF
L1 10uH
C4 0.1uF
C3 0.1uF
(PCINT6/XTAL1/TOSC1) PB6
(PCINT7/XTAL2/TOSC2) PB7
(PCINT21/OC0B/T1) PD5
(PCINT22/OC0A/AIN0) PD6
ADC6
ADC7
PC0 (ADC0/PCINT8)
PC1 (ADC1/PCINT9)
PC2 (ADC2/PCINT10)
PC3 (ADC3/PCINT11)
PC4 (ADC4/SDA/PCINT12)
PC5 (ADC5/SCL/PCINT13)
PC6 (RESET/PCINT14)
(PCINT3/OC2A/MOSI) PB3
(PCINT2/SS/OC1B) PB2
(PCINT4/MISO) PB4
PB5 (SCK/PCINT5)
PD0 (RXD/PCINT16)
PD1 (TXD/PCINT17)
(PCINT19/OC2B/INT1) PD3
(PCINT20/XCK/T0) PD4
(PCINT23/AIN1) PD7
(PCINT0/CLKO/ICP1) PB0
(PCINT1/OC1A) PB1
PD2 (INT0/PCINT18)
GND
ATMEGA328P-AU
MOSFET3
INDICATION LEDS
LED1 R11 3.3K LED1
D11/MOSI R12 3.3K LED2
LED3 R13 3.3K LED3
RESET SWITCH
R6 10K
SW1
C9 0.1uF
CONTROL SWITCH
SWITCH
R10 10K
SW2
MH1
MH2
POWER SIGNALS
VCC_5V
GND

MOS_4 MOS_4

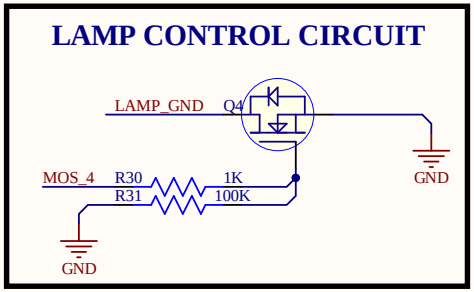
DECOUPLING CAPACITOR



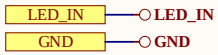
LAMP



LAMP CONTROL CIRCUIT



POWER SIGNALS



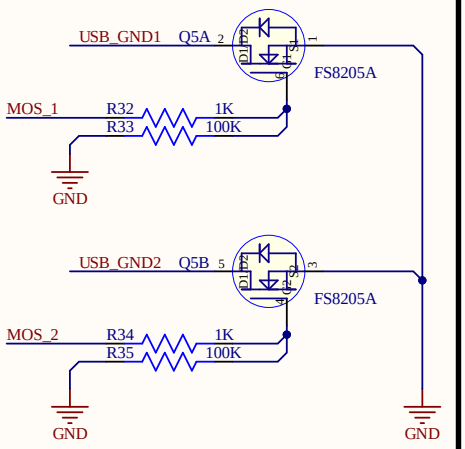
MOSFET CONTROL SIGNALS



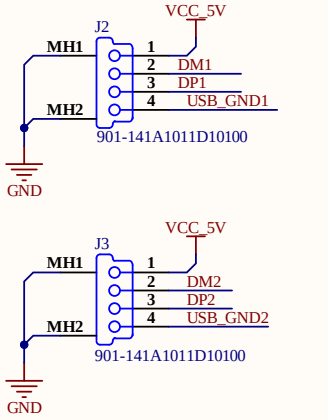
USB COMMUNICATION



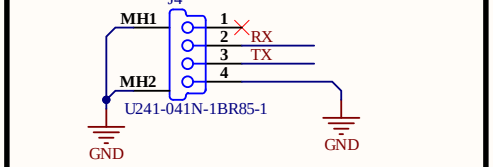
MOSFET CONTROL CIRCUIT



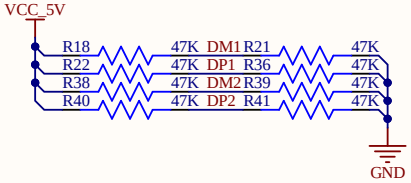
CHARGER OUTPUT



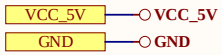
EXTERNAL COMMUNIICATION

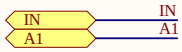


RESISTOR DIVIDER FOR USB DATA LINES



POWER SIGNALS

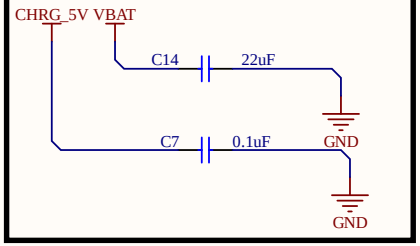




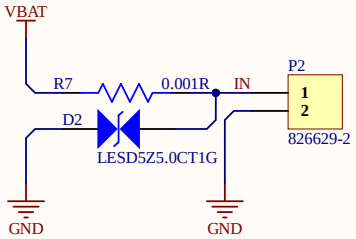
ISP SIGNALS



DECOUPLING CAPACITOR

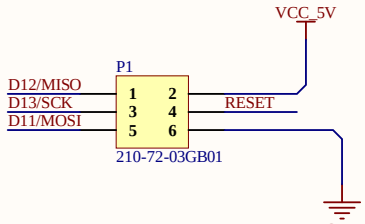


BATTERY HEADER

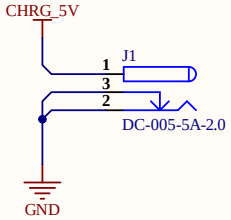


IN signal comes from Current sense amplifier circuitry

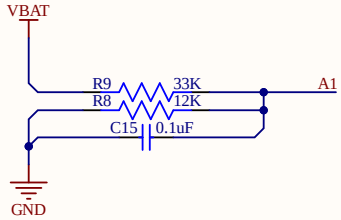
ISP PROGRAMMING HEADER



DC POWER HEADER

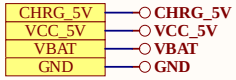


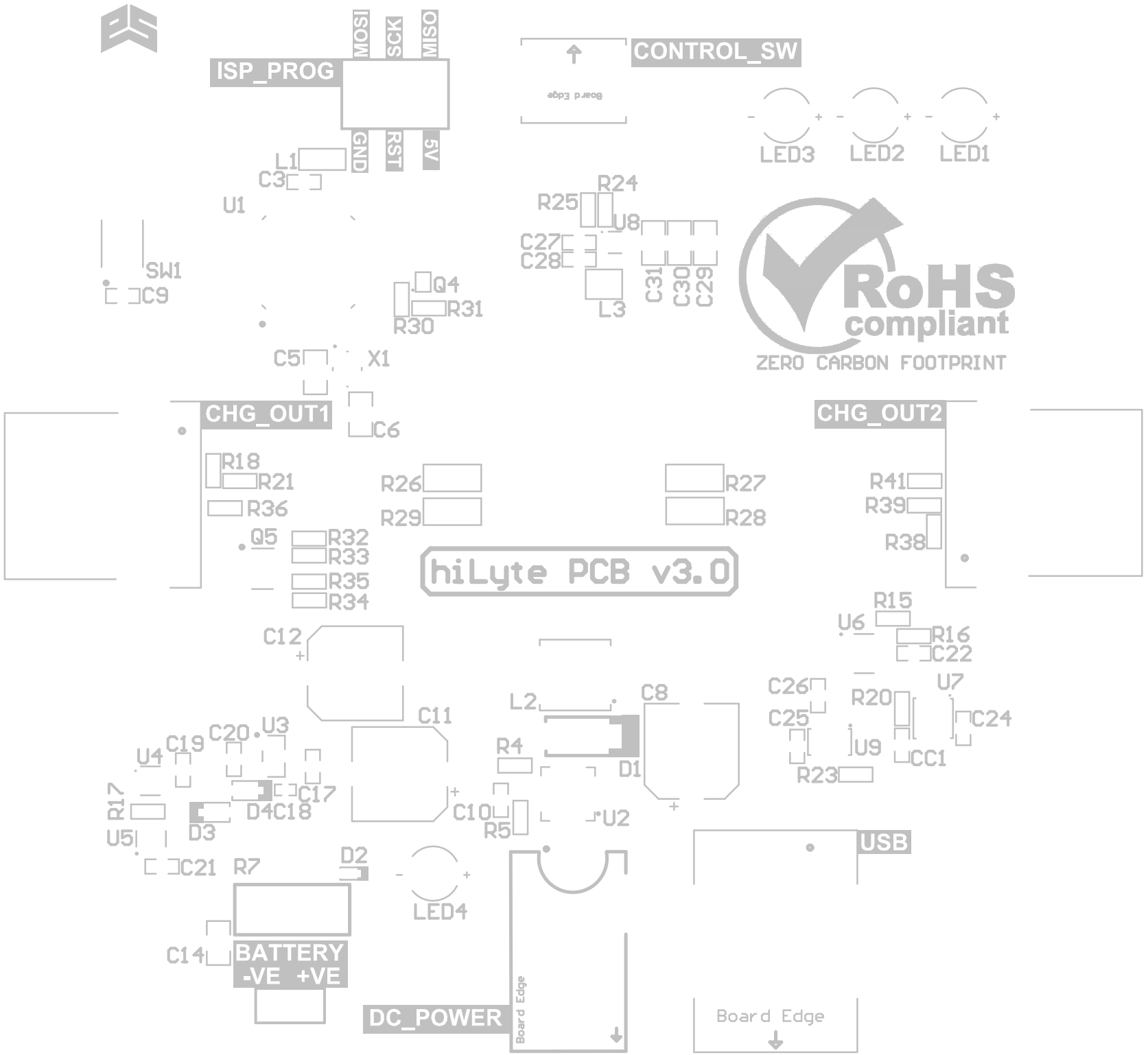
FEEDBACK NETWORK



A1 signal comes from Controller

POWER SIGNALS




ISP_PROG
MOSI
SCK
MISO
GND
RST
5V
CONTROL_SW
Board Edge
LED3
LED2
LED1
L1
C3
U1
SW1
C9
Q4
R31
R30
R24
R25
U8
C27
C28
C31
C30
C29
L3
RoHS
compliant
ZERO CARBON FOOTPRINT
C5
X1
C6
CHG_OUT1
CHG_OUT2
R18
R21
R36
R26
R29
R27
R28
R41
R39
R38
Q5
R32
R33
R35
R34
hiLyte PCB v3.0
R15
U6
R16
C22
C12
C11
L2
C8
C26
C25
R20
U7
C24
U9
CC1
R23
U4
C19
C20
U3
R4
D1
R17
D4
C18
C17
C10
R5
U2
D3
U5
C21
R7
D2
LED4
USB
C14
BATTERY
-VE +VE
DC_POWER
Board Edge
Board Edge